Spring 2025 Astronomy 98: The Relationship Between Mass, Radius, and Gravitational Acceleration of Black Holes

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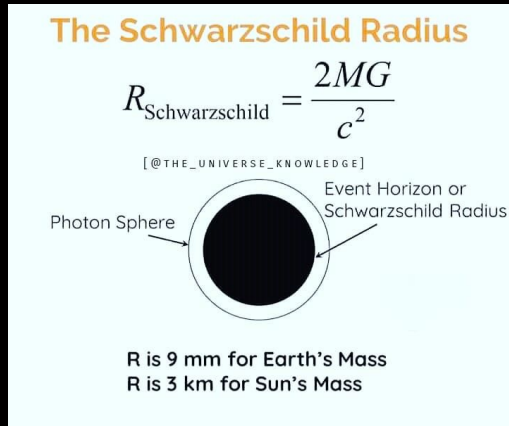
Overview

The purpose of this project is to demonstrate the relationship graphically between the mass of black holes, the radius of event horizon, and the gravitational acceleration at that point.

I will be using Newton's Gravitational Force Equation and Schwarzschild's Radius equation.

$$F = G\frac{m_1 m_2}{r^2}$$



Methods

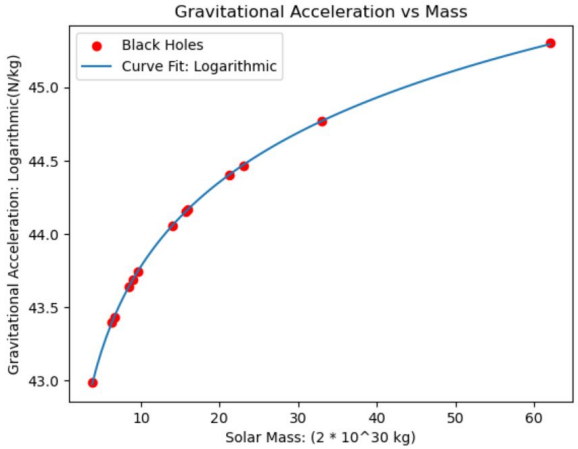
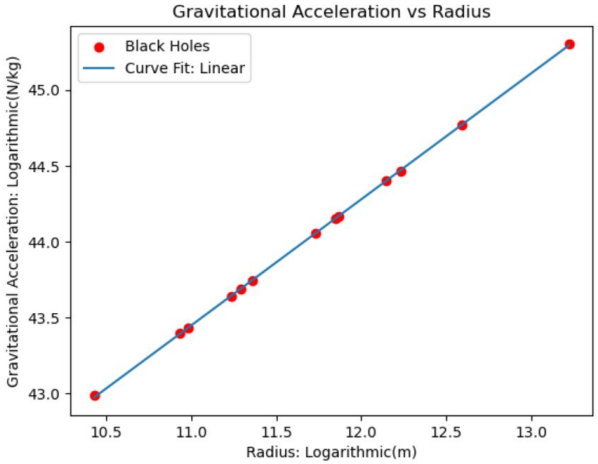
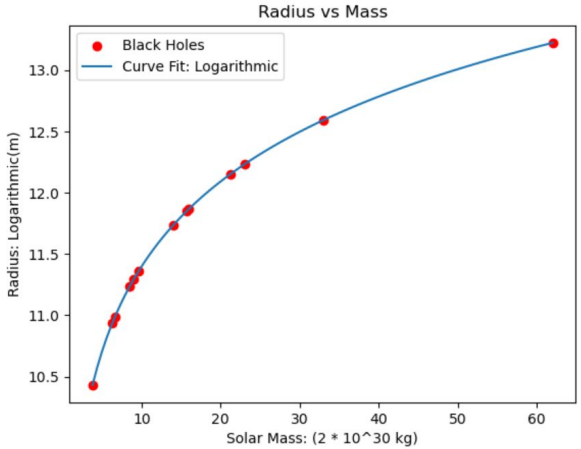
Create data set of black hole masses (Wikipedia and Nasa)

Plug data into equations, and then add new columns for radius and gravitational acceleration.

Plot the following graphs: Gravitational Acceleration vs Mass, Gravitational Acceleration vs Radius, Radius vs Mass

Results

Note: Some of the data needed to be on a logarithmic scale so that it can actually fit.

	Stellar Black Holes	Solar Mass	Radius (Logarithmic: m) \
0	Gaia BH1	9.62	11.358930
1	Cygnus X-1	21.20	12.149087
2	V404 Cygni	9.00	11.292310
3	GRP J1655-40	6.30	10.935635
4	A0620-00	6.60	10.982155
5	GW150914	62.00	13.222220
6	Gaia BH3	33.00	12.591593
7	IC 10 X-1	23.00	12.230580
8	M33 X-7	15.70	11.848746
9	GRS 1915+105	14.00	11.734143
10	SS 433	16.00	11.867674
11	XTE J1650-500	3.80	10.430087
12	GS 2000+25	8.50	11.235152

	Gravitational Acceleration (Logarithmic: N/kg)
0	43.744364
1	44.400021
2	43.689509
3	43.397024
4	43.435054
5	45.303865
6	44.770977
7	44.468144
8	44.149745
9	44.054580
10	44.165480
11	42.986140
12	43.642500